

UGA Mock-1, Solution Key

Maximum Marks: 120

Duration: 120 Minutes

Each correct answer carries 4 marks, unattempted question carries 1 mark, wrong answer carries 0 mark

Q. 1 To determine the properties of the symmetric idempotent matrix A , let $A = [a_{ij}]$. The matrix is idempotent, meaning $A^2 = A$, and symmetric, meaning $a_{ij} = a_{ji}$. Consider the i -th diagonal element of A^2 , which is computed by taking the dot product of the i -th row and i -th column of A . Because A is symmetric, this dot product is $\sum_j a_{ij}a_{ji} = \sum_j a_{ij}^2 = a_{ii}^2 + \sum_{j \neq i} a_{ij}^2$. Since $A^2 = A$, this must equal a_{ii} . Thus, we have the algebraic equation $a_{ii}^2 + \sum_{j \neq i} a_{ij}^2 = a_{ii}$. Rearranging this gives $a_{ii} - a_{ii}^2 = \sum_{j \neq i} a_{ij}^2$. As a sum of squared real numbers is always non-negative, meaning $\sum_{j \neq i} a_{ij}^2 \geq 0$. Therefore, $a_{ii} - a_{ii}^2 \geq 0$. Factoring this yields $a_{ii}(1 - a_{ii}) \geq 0$. Solving this basic quadratic inequality gives $0 \leq a_{ii} \leq 1$, all diagonal elements lie between 0 and 1. **Option (c) is correct.**

Q. 2 We evaluate the limit of the sum of $\frac{1}{\binom{n}{k}}$. For large n , the terms are $1 + \frac{1}{n} + \dots + \frac{1}{n} + 1$. The intermediate terms for $2 \leq k \leq n-2$ are bounded above by the second term, meaning their sum approaches 0 as $n \rightarrow \infty$. Thus, the limit is precisely $1 + 0 + 1 = 2$. **Option (c) is correct.**

Q. 3 Let,

$$z_1 \cdot \overline{z_1} = z_2 \cdot \overline{z_2} = z_3 \cdot \overline{z_3} = r^2 > 0,$$

and notice that,

$$z_1 z_2 + z_3 \in \mathbb{R} \text{ if and only if } z_1 z_2 + z_3 = \overline{z_1} \cdot \overline{z_2} + \overline{z_3}.$$

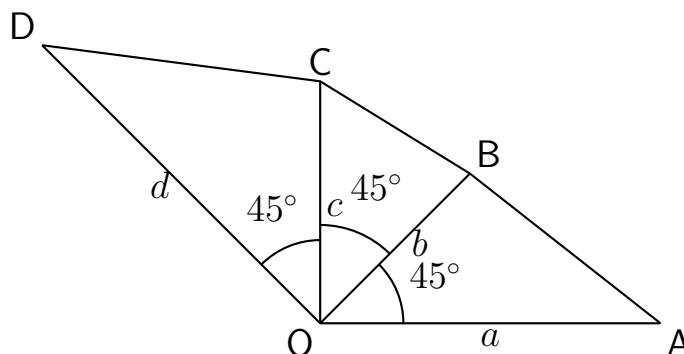
Then,

$$\begin{aligned} \frac{r^2}{z_1 z_2 z_3} &= \frac{z_1 z_2 + z_3}{z_1 z_2 + r^2 z_3} = \frac{z_1 z_3 + z_2}{z_1 z_3 + r^2 z_2} = \frac{z_2 z_3 + z_1}{z_2 z_3 + r^2 z_1} \\ \frac{(z_1 - 1)(z_2 - z_3)}{(z_2 - z_3)(z_1 - r^2)} &= \frac{z_1 - 1}{z_1 - r^2} = \frac{z_2 - 1}{z_2 - r^2} = \frac{z_3 - 1}{z_3 - r^2} = \frac{z_1 - z_2}{z_1 - z_2} = 1. \end{aligned}$$

Hence $z_1 z_2 z_3 = r^2$ and consequently $r^3 = r^2$. Therefore $r = 1$ and $z_1 z_2 z_3 = 1$. Hence, **option (a) is correct.**

Q. 4 To find the remainder of $\binom{100}{50}$ modulo 7, we apply Lucas' Theorem. First, convert to base 7: $100 = (202)_7$ and $50 = (101)_7$. Therefore, $\binom{100}{50} \equiv \binom{2}{1} \binom{0}{0} \binom{2}{1} = 2 \times 1 \times 2 = 4 \pmod{7}$. **Option (d) is correct.**

Q. 5 The given equation is the equality case of the triangle inequality for lengths of sides of a quadrilateral at three 45° angles from one single vertex (O). The sum of the segment lengths equals the total distance AD , making the four endpoints collinear. Using area addition for the sub-triangles, $\frac{1}{2}ad \sin 135^\circ = \frac{1}{\sqrt{2}}(ab + bc + cd)$, which simplifies to $ad = ab + bc + cd$.



Therefore, $\frac{(a+c)(b+d)}{ad} = \frac{ab+bc+cd+ad}{ad} = \frac{ad+ad}{ad} = 2$.

Option (a) is correct.

Q. 6 Given the cyclic fractions for p and q , algebraic factorization yields $|p - q| = \frac{|(a-b)(b-c)(c-a)|}{abc}$. By the triangle inequality for triangle sides, $|a - b| < c$, $|b - c| < a$, and $|c - a| < b$. The product is strictly less than abc , meaning $|p - q| < 1$. **Option (d) is correct.**

Q. 7 The expressions $p^2 + q^3$ and $p^3 + q^2$ are both primes. To avoid both being even composite numbers, one prime must be 2. Let $p = 2$. Testing $q = 3$ yields 31 and 17, which are both prime. Any other prime q introduces a factor of 3. The requested sum evaluates to $2 + 3 + 4 + 9 + 8 + 27 = 53$. **Option (a) is correct.**

Q. 8 We analyze the functional equation $f(x)f(f(x)) = 1$. Substituting $y = f(x)$, we deduce $f(y) = 1/y$ for all y in the range. Since $f(1000) = 999$, the continuous function's range covers $[1/999, 999]$, guaranteeing 500 is in the range (by the IVT). Hence, **Option (c) is correct.**

Q. 9 Using the base change rule for logarithms, $x = \frac{\ln 2}{\ln 3 + \ln 5}$, $y = \frac{\ln 3}{\ln 2 + \ln 5}$, and $z = \frac{\ln 5}{\ln 2 + \ln 3}$. This exact symmetric structure identically satisfies the algebraic relation $xy + yz + zx + 2xyz = 1$. **Option (b) is correct.**

Q. 10 The arithmetic sequence from 1 to 551 has exactly 56 elements. We group them into 28 unique pairs that each sum to 552. To construct a subset without any such pairs, we can select at most one element from each pair, giving a maximum of 28 elements. **Option (d) is correct.**

Q. 11 We evaluate the definite integral with bounds 0 and 4. Substituting $x - 2 = 2 \sin \theta$ shifts the integration bounds symmetrically from $-\pi/2$ to $\pi/2$. The properties of logarithms cause the resulting symmetric components to perfectly cancel out to 0. **Option (d) is correct.**

Q. 12 We are given the polynomial $P(x) = (1 + x + x^2 + x^3 + x^4 + x^5 + x^6 + x^7)^2 - x^7$

Let,

$$\begin{aligned} 1 + x + x^2 + \cdots + x^6 &= a \\ \Rightarrow \frac{x^7 - 1}{x - 1} &= a \\ \Rightarrow x^7 &= a(x - 1) + 1. \end{aligned}$$

Hence,

$$\begin{aligned} P(x) &= (a + a(x - 1) + 1)^2 - a(x - 1) - 1 \\ &= (ax + 1)^2 - ax + a - 1 \\ &= a^2x^2 + 2ax + 1 - ax + a - 1 \\ &= a^2x^2 + ax + a \\ &= a(ax^2 + x + 1) \\ &= (1 + x + x^2 + \cdots + x^6)(1 + x + \cdots + x^6 + x^7 + x^8) \end{aligned}$$

Hence, **option (d) is correct.**

Q. 13 For the moving point P , set $A = (-a, 0)$ and $B = (a, 0)$. Expanding $PA^2 - PB^2 = \text{const}$ geometrically resolves to $4ax = \text{const}$, identifying the locus as a vertical line perpendicular to the segment AB . **Option (c) is correct.**

Q. 14 The given discrete sum converges to the definite integral $\int_0^1 \sqrt{1 - x^2} dx = \pi/4 \approx 0.785$. As a decreasing function, the finite limit strictly remains below the 0.79 threshold. **Option (a) is correct.**

- Q. 15** Finding the domains of the two radical functions gives a restricted valid overlap of $x \in [4, 6]$. The maximum difference occurs precisely at the boundary $x = 4$, giving $\sqrt{12} - 0 = 2\sqrt{3}$. **Option (b) is correct.**
- Q. 16** In the right triangle, placing the origin at the right angle B defines distance constraints for the partitioned hypotenuse points D and E . Applying the distance formula, we get $BD^2 = 9$ and $BE^2 = 16$. Summing these, we get the relationship for hypotenuse as $AB^2 + BC^2 = 45$, so $AC = 3\sqrt{5}$. **Option (c) is correct.**
- Q. 17** Define $f(t) = t + t^3 + \sin t$. Then $f'(t) > 0$ for all real t , thus implying that f is strictly increasing (hence, one to one). The given functional equation $f(x) = f(x^2)$ confirms that we must have $x = x^2$, leaving exactly two real solutions: 0 and 1. **Option (c) is correct.**
- Q. 18** We test integers modulo 3 in the greatest integer function $[\frac{n^2}{3}]$. Generating prime outcomes requires isolating specific moduli cases, which exclusively yield the primes 3 (when $k = 1$ for $3k$) and 5 (when $k = 1$ for $3k + 1$). Thus, exactly 2 primes are produced. **Option (b) is correct.**
- Q. 19** The equation $3\sin \alpha + 4\cos \alpha - 7\cos \beta = 12$ forces each trigonometric component to its theoretical extreme, since $\max(3\sin \alpha + 4\cos \alpha) = 5$, and $\max(-7\cos \beta) = 7$. The maxima require $\cos \beta = -1$ and $\cos \alpha = 4/5$. Evaluating the expression with these values results in exactly 24. **Option (c) is correct.**
- Q. 20** Note that the integrands in I_1 and I_2 are strict inverse functions of each other mapping the unit interval $[0, 1]$ onto itself. By symmetric geometric properties over the unit square, their areas are identically equal, so $I_1 = I_2$. **Option (c) is correct.**
- Q. 21** The maximum area for a quadrilateral with side lengths a, b, c, d depends on the sin of its angles. To achieve the equality $\frac{1}{2}ab\sin \theta_1 + \frac{1}{2}cd\sin \theta_2 = \frac{1}{2}ab + \frac{1}{2}cd$ requires opposite angles (θ_1 and θ_2) each to be equal to 90° , resulting to sum 180° , making it cyclic. **Option (d) is correct.**
- Q. 22** For a function to be idempotent, $f(f(x)) = f(x)$ must hold. One must choose k fixed points in $\binom{n}{k}$ ways, while the remaining $n - k$ elements must map exclusively into these k points (k^{n-k} ways). The total summation is $\sum_{k=1}^n \binom{n}{k} k^{n-k}$. **Option (a) is correct.**
- Q. 23** We construct a comparison function $g(x) = e^x + x$ to match the integral values. Expanding the quadratic difference $\int_0^1 (f(x) - g(x))^2 dx$ perfectly evaluates to 0, confirming that $f(x) = e^x + x$. Thus, $f(0) = 1$. **Option (b) is correct.**
- Q. 24** Red and white balls drawn without replacement. Creating the baseline denominator determines the total possible indiscriminate pairings is $\frac{(2n)!}{(2^n n!)}$. Selecting only successful red-white matched pairs restricts the favorable outcomes to $n!$, generating a final probability of $\frac{2^n}{\binom{2n}{n}}$. **Option (a) is correct.**

Q. 25 We have,

$$\begin{aligned}
\frac{x^2}{x+1} + \frac{y^2}{y+2} + \frac{z^2}{z+3} &= x + y + z + 106 \\
\Rightarrow \left(\frac{x^2}{x+1} - x \right) + \left(\frac{y^2}{y+2} - y \right) + \left(\frac{z^2}{z+3} - z \right) &= 106 \\
\Rightarrow \frac{-x}{x+1} + \frac{-2y}{y+2} + \frac{-3z}{z+3} &= 106 \\
\Rightarrow \left(\frac{-x}{x+1} + 1 \right) + \left(\frac{-2y}{y+2} + 2 \right) + \left(\frac{-3z}{z+3} + 3 \right) &= 112 \\
\Rightarrow \frac{1}{x+1} + \frac{4}{y+2} + \frac{9}{z+3} &= 112 \\
\Rightarrow 4 \left(\frac{1}{4x+4} + \frac{1}{y+2} + \frac{9}{4z+12} \right) &= 112 \\
\Rightarrow \frac{1}{4x+4} + \frac{1}{y+2} + \frac{9}{4z+12} &= 28
\end{aligned}$$

Option (d) is correct.

Q. 26 Suppose such a continuous function f exists. We are given that, $f(f(x)) + f(x) + x = 0$. Putting $f(x)$ in place of x , we get,

$$f(f(f(x))) + \cancel{f(f(x))} + \cancel{f(x)} = 0 = \cancel{f(f(x))} + \cancel{f(x)} + x \Rightarrow f(f(f(x))) = x$$

We shall show that f is one-one. To this point, let $f(x) = f(y)$

Then,

$$f(f(x)) = f(f(y)) \Rightarrow f(f(f(x))) = f(f(f(y))) \Rightarrow x = y.$$

Hence f is one-one, and also continuous. So f is strictly monotone. If possible, let f be strictly decreasing. Then

$$x > y \Rightarrow f(x) < f(y) \Rightarrow f(f(x)) > f(f(y)) \Rightarrow f(f(f(x))) < f(f(f(y))) \Rightarrow x < y, \text{ contradiction!}$$

So, f must be strictly increasing.

Then, for $x > y$, we have

$$f(x) > f(y) \Rightarrow f(f(x)) > f(f(y))$$

Therefore,

$$0 = x + f(x) + f(f(x)) < y + f(y) + f(f(y)) = 0, \text{ contradiction!}$$

Hence no such function exists. **Option (c) is correct.**

Q. 27 The orthogonal matrices A and B have determinants of ± 1 . For their sum to be 0, they must carry opposite determinant signs. Therefore $\det(A) \cdot \det(B) = -1$. Now, $\det(A+B) = \det(A(B^{-1}+A^{-1})B) = \det(A(B^\top+A^\top)B) = \det(A) \det((A+B)^\top) \det(B) = \det(A) \det(B) \det(A+B) = -\det(A+B)$. Hence, $\det(A+B) = 0$. **Option (b) is correct.**

Q. 28 For z_1, z_2, z_3 as the vertices of an equilateral triangle, we have $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$. And z_0 being the centroid, $z_1 + z_2 + z_3 = 3z_0 \Rightarrow (z_1 + z_2 + z_3)^2 = 9z_0^2 \Rightarrow z_1^2 + z_2^2 + z_3^2 + 2(z_1 z_2 + z_2 z_3 + z_3 z_1) = 9z_0^2 \Rightarrow 3(z_1^2 + z_2^2 + z_3^2) = 9z_0^2 \Rightarrow \frac{z_1^2 + z_2^2 + z_3^2}{z_0^2} = 3$. **Option (d) is correct.**

Q. 29 Letting $I(a) = \int_0^1 \frac{x^a - 1}{\ln x} dx$, we get

$$\begin{aligned} I'(a) &= \int_0^1 x^a dx = \frac{1}{a+1} \\ \Rightarrow I(a) &= \int \frac{da}{a+1} = \ln|a+1| + c \end{aligned}$$

Now, $I(0) = 0 \Rightarrow c = 0 \Rightarrow I(a) = \ln|a+1|$. So, the given integral

$$\begin{aligned} &= \int_0^1 \frac{x^{2027} - x^3}{\ln x} dx \\ &= \int_0^1 \frac{x^{2027} - 1}{\ln x} dx - \int_0^1 \frac{x^3 - 1}{\ln x} dx \\ &= \ln(1 + 2027) - \ln(1 + 3) \\ &= \ln\left(\frac{2028}{4}\right) \\ &= \ln 507. \end{aligned}$$

Option (b) is correct.

Q. 30 Integrating the given inequality $f(x)f'(x) \geq \cos x$ bounds the squared function, resulting in $f^2(x) \geq f^2(0) + 2 \sin x$. As $\sin$ is an oscillating function, it has no limit as $x \rightarrow \infty$. Hence the limit of $f(x)$ as $x \rightarrow \infty$ also does not exist. **Option (d) is correct.**

-End of the Solution Key-